Ans. For 5: 10. Expression: $2(x+3)-6=2x$: the result is always twice the chosen number.
Link A09: algebraic notation proves what numerical trials suggest.

Sheet 12 Tue 17 Nov. 2026 *Intersection of two lines; Python collinearity function*

1. Warm-up · Quick [EQ] Solve $2x+1=-x+7$.
   Ans. $3x=6$, so $x=2$.
2. Core · Method [GEO] Coordinates of the intersection of $y=2x+1$ and $y=-x+7$.
   Ans. $x=2$ then $y=2\times 2+1=5$: $(2;5)$.
   Check in the second equation: $-2+7=5$.
3. Core + · Code reading [ALG] What do `alignes(0,0,1,2,2,4)` and `alignes(0,0,1,2,2,5)` return?
   ```
   def alignes(xa, ya, xb, yb, xc, yc):
       return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
   ```
   Ans. `True` then `False`: determinants $1\times 4-2\times 2=0$ and $1\times 5-2\times 2=1\neq 0$.
4. Review · Expand [DEV] Expand and simplify $3(x-2)$.
   Ans. $3x-6$.
5. Thread · Prove [LIT] Show that the sum of three consecutive integers is a multiple of 3.
   Ans. With $n-1$, n, $n+1$: sum $=3n$, a multiple of 3.
   Choosing the middle term simplifies the calculation.

Real numbers and intervals

Sheet 13 Thu 19 Nov. 2026 *Sets of numbers: $\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$*

1. Warm-up · Quick [NUM] Give the decimal form of $\frac{3}{8}$ and an approximation of $\frac{5}{6}$ to within 10^{-3}.
   Ans. 0.375; 0.833.
2. Core · Method [NUM] Give the smallest set ($\mathbb{N},\mathbb{Z},\mathbb{D},\mathbb{Q},\mathbb{R}$) containing each of these numbers: -4; 0.25; $\frac{1}{3}$; $\sqrt{2}$; $\frac{6}{3}$.
   Ans. $-4\in\mathbb{Z}$; $0.25\in\mathbb{D}$; $\frac{1}{3}\in\mathbb{Q}$; $\sqrt{2}\in\mathbb{R}$; $\frac{6}{3}=2\in\mathbb{N}$.
   The smallest set is the key: $\frac{6}{3}$ is a natural number.
3. Core + · True/false [NUM] True or false? Justify. The number $\frac{1}{3}$ is a decimal number.
   Ans. False. If there were integers a and n with $\frac{1}{3}=\frac{a}{10^n}$, then $10^n=3a$ would be divisible by 3: impossible (the sum of the digits of 10^n is 1).
4. Review · Quick [SUB] Calculate $(x+1)(x-3)$ for $x=2$.
   Ans. -3.
5. Thread · Quick [SOL] How many faces, vertices and edges does a right triangular prism have?
   Ans. 5 faces (2 triangles, 3 rectangles), 6 vertices, 9 edges.

Sheet 14 Mon 23 Nov. 2026 *Decimal bounds and rounding*

1. Warm-up · Quick [NUM] Round 3.14159 to the nearest 10^{-2} and then to the nearest 10^{-3}.
   Ans. 3.14 then 3.142.
   The next digit decides: $1<5$ then $5\geqslant 5$.
2. Core · Method [NUM] Bound $\sqrt{2}$ between two decimals that differ by 10^{-2}.
   Ans. $1.41^2=1.9881<2<2.0164=1.42^2$: $1.41<\sqrt{2}<1.42$.
3. Core + · Code reading [ALG] What is the final value of `k`? What can be deduced about $\sqrt{2}$?
   ```
   k = 10
   ```

```
while (k/10)**2 < 2:
    k = k + 1
```

Ans. $k = 15$: $1.4^2 = 1.96 < 2$ and $1.5^2 = 2.25 \geqslant 2$, so $1.4 < \sqrt{2} \leqslant 1.5$.

4. Review · Quick [QP] 5 kg of rice cost 12 500 Ar. How much do 8 kg cost?

Ans. $\dfrac{12\,500}{5} \times 8 = 20\,000$ Ar.

5. Thread · True/false [DEM] True or false? Justify. A quadrilateral whose diagonals have equal length is a rectangle.

Ans. False. Counter-example: an isosceles trapezium has diagonals of equal length without being a rectangle.

Sheet 15 Tue 24 Nov. 2026 *Irrational numbers: $\frac{1}{3}$ is not decimal, $\sqrt{2}$ is irrational*

1. Warm-up · Quick [CAL] Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b the smallest possible integer.

Ans. $\sqrt{12} = 2\sqrt{3}$; $\sqrt{50} = 5\sqrt{2}$.

2. Core · Sorting [NUM] Among $0.333\ldots$; $\sqrt{9}$; $\sqrt{2}$; π; $\dfrac{22}{7}$, which are rational?

Ans. $0.333\ldots = \dfrac{1}{3}$; $\sqrt{9} = 3$; $\dfrac{22}{7}$: rational. $\sqrt{2}$ and π are irrational.

$\frac{22}{7}$ is only an approximation of π, not equal to π.

3. Core + · Reasoning [DEM] Suppose $\sqrt{2} = \dfrac{p}{q}$ with $\dfrac{p}{q}$ in lowest terms, so $p^2 = 2q^2$. What can be said about the parity of p, then of q? What contradiction arises?

Ans. p^2 even, so p even, $p = 2k$; then $4k^2 = 2q^2$, $q^2 = 2k^2$: q even. Contradiction with the fraction being in lowest terms.

4. Review · Table [STA] Values 1 ; 2 ; 3 ; 4 with frequencies 5 ; 8 ; 4 ; 3. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Frequency	5	8	4	3

Ans. Total 20; relative frequency $\dfrac{8}{20} = 40\,\%$; mean $\dfrac{5+16+12+12}{20} = 2.25$.

5. Thread · Quick [REP] Give the coordinates of the reflection of $A(3;-2)$ in the y-axis.

Ans. $(-3;-2)$.

Pitfall: changing the sign of the ordinate (reflection in the x-axis).

Sheet 16 Thu 26 Nov. 2026 *Intervals of $\mathbb{R}$: notation and representation*

1. Warm-up · Quick [CMP] Compare -3 and -5, then $-\dfrac{1}{2}$ and -0.6.

Ans. $-3 > -5$; $-0.5 > -0.6$.

For negative numbers, the one with the smaller absolute value is the greater.

2. Core · Method [NUM] Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.

Ans. $[-2;5[$; $]3;+\infty[$; $]-\infty;0]$.

The bracket faces inwards if the endpoint is included; ∞ is always excluded.

3. Core + · MCQ [NUM] The intersection $]-\infty;2[\cap [-1;4]$ is:

A. $]2;4]$ **B.** $]-\infty;4]$ **C.** $[-1;2[$ **D.** $[-1;2]$

Ans. C. $[-1;2[$

Intersection: values satisfying both conditions $x < 2$ and $-1 \leqslant x \leqslant 4$.

4. Review · Table [LEC] Journey times (min).

Class	$[0;10[$	$[10;20[$	$[20;30[$
Frequency	4	9	7